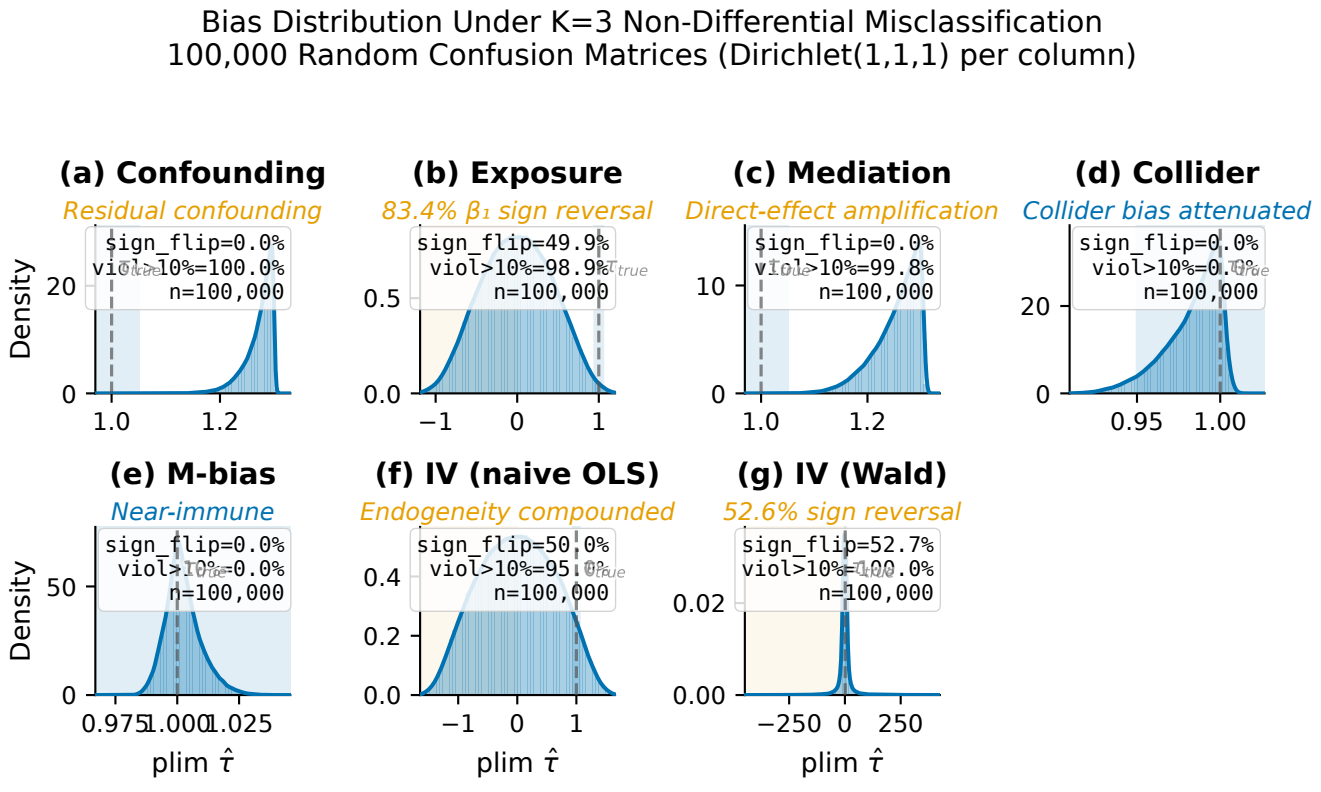

Bias Distribution Under K=3 Non-Differential Misclassification
100,000 Random Confusion Matrices (Dirichlet(1,1,1) per column)
(a) Confounding
Residual confounding
sign_flip=0.0%
viol>10%=100.0%
n=100,000
(b) Exposure
83.4% β₁ sign reversal
sign_flip=49.9%
viol>10%=98.9%
n=100,000
(c) Mediation
Direct-effect amplification
sign_flip=0.0%
viol>10%=99.8%
n=100,000
(d) Collider
Collider bias attenuated
sign_flip=0.0%
viol>10%=0.0%
n=100,000
(e) M-bias
Near-immune
sign_flip=0.0%
viol>10%=0.0%
n=100,000
(f) IV (naive OLS)
Endogeneity compounded
sign_flip=50.0%
viol>10%=95.0%
n=100,000
(g) IV (Wald)
52.6% sign reversal
sign_flip=52.7%
viol>10%=100.0%
n=100,000
Density
plim τ̂